

$$Z = BX + (1-B)Y$$

$$X, Y \text{ esponenziali e indep} \dots \dots f_X(x) = e^{-x} u(x)$$

$$E[X] = 1 \quad E[Y] = \frac{1}{2}$$

$$f_Y(y) = 2e^{-2y} u(y)$$

$$B \in \{0, 1\}, \text{ indep da } X \text{ e } Y, \quad P[B=1] = d$$

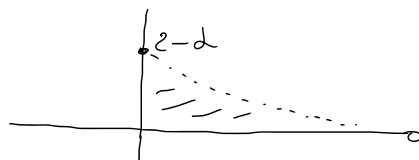
$$Z|_{B=0} = Y$$

$$f_Z(z) = P[B=0] f_{Z|B=0}(z) + P[B=1] f_{Z|B=1}(z)$$

$$Z|_{B=1} = X$$

$$f_Z(z) = (1-d) 2e^{-2z} u(z) + d e^{-z} u(z) = [d e^{-z} + (1-d) 2e^{-2z}] u(z)$$

$$\int_0^{+\infty} (d e^{-z} + (1-d) 2e^{-2z}) dz$$



$$= \left[-d e^{-z} - (1-d) e^{-2z} \right]_0^{+\infty} = d + (1-d) = 1$$

$$E[g(Z)] = \int_{-\infty}^{+\infty} g(z) f_Z(z) dz = \int_{-\infty}^{+\infty} e^{-sz} [d e^{-z} + (1-d) 2e^{-2z}] u(z) dz$$

$$g(x) = e^{-sx} \quad s > 0 \quad = \int_0^{+\infty} [d e^{-(s+1)z} + (1-d) 2e^{-(s+2)z}] dz$$

$$= \left[-d \frac{e^{-(s+1)z}}{s+1} - (1-d) 2 \frac{e^{-(s+2)z}}{s+2} \right]_0^{+\infty} = \frac{d}{s+1} + \frac{2(1-d)}{s+2} = \phi_Z(s)$$

$$E[g(Z)|B=0] = E[g(Y)|B=0] = E[g(Y)] = \int_{-\infty}^{+\infty} g(y) f_Y(y) dy$$

$$= \int_{-\infty}^{+\infty} e^{-sy} 2e^{-2y} u(y) dy = \int_0^{+\infty} 2e^{-(s+2)y} dy = \left[-2 \frac{e^{-(s+2)y}}{s+2} \right]_{y=0}^{y \rightarrow \infty} = \frac{2}{s+2}$$

$$E[g(Z)|B=1] = E[g(X)|B=1] = E[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx$$

$$= \int_{-\infty}^{+\infty} e^{-sx} e^{-x} dx = \int_0^{+\infty} e^{-(s+1)x} dx = \frac{1}{s+1}$$

$$= \int_{-\infty}^{+\infty} e^{-sx} e^{-x} u(x) dx = \int_0^{+\infty} e^{-(s+1)x} dx = \left[-\frac{e^{-(s+1)x}}{s+1} \right]_0^{+\infty} = \frac{1}{s+1}$$

$$E[g(z)] = P(B=0) E[g(z)|B=0] + P(B=1) E[g(z)|B=1]$$

$$= (1-d) \frac{2}{s+2} + d \frac{1}{s+1} \quad \frac{d}{ds} \quad 2(1-d)(-1)(s+2)^{-2} + d(-1)(s+1)^{-2}$$

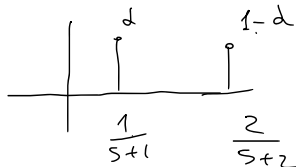
$$\phi_z(0) = \lim_{s \rightarrow 0} \phi_z(s) = \frac{2(1-d)}{2} + d = 1 \quad = \frac{-2(1-d)}{(s+2)^2} - \frac{d}{(s+1)^2} \xrightarrow{s \rightarrow 0} \frac{-2(1-d)}{4} - d$$

$$\phi_z(0) = E[e^{-sz}]_{s=0} = E[1] = 1 \quad = -\frac{1}{2} + \frac{d}{2} - d = -\frac{1}{2} - \frac{d}{2} = -\frac{1+d}{2}$$

$$\begin{aligned} \frac{d}{ds} \phi_z(s) \Big|_{s=0} &= \lim_{s \rightarrow 0^+} \frac{(1-d) \frac{2}{s+2} + \frac{d}{s+1} - 1}{s} = \frac{2(1-d)(s+1) + d(s+2) - s(s+2)(s+1)}{s(s+2)(s+1)} \\ &= \frac{[2s - 2d s + 2 - 2d + d s + 2d] - [s^3 + 3s^2 + 2s]}{s(s+2)(s+1)} = \frac{[2s - d s + 2]}{s(s+2)(s+1)} \end{aligned}$$

$$= \frac{-s - s^2 - d s}{s(s+2)(s+1)} = \frac{-d - d - s}{(s+2)(s+1)} \xrightarrow{s \rightarrow 0} \frac{-1-d}{2}$$

$$C(B) = E[e^{-z} | B]$$



$$1-X \dots a=-1, b=1$$

$$Y = aX + b \dots g(x) = ax + b$$

$$g(x) = y \Leftrightarrow ax + b = y \Leftrightarrow x = \frac{y-b}{a} \quad a \neq 0$$

$$Y = aX$$

$$x'_j(y) = \frac{1}{a}$$

$$|x'(y)| = \frac{1}{|a|}$$

$$Y = X + b$$

$$f_Y(y) = |x'(y)| f_X(x_1(y)) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

$$f_{aX+b}(u) = \frac{1}{|a|} f_X\left(\frac{u-b}{a}\right) \begin{matrix} \xrightarrow{b=0} \frac{1}{|a|} f_X\left(\frac{u}{a}\right) \\ \xrightarrow{a=1} f_X(u-b) \end{matrix} = \frac{1}{|a|} f_X\left(\frac{y-1}{-1}\right) = f_X(1-y)$$

$$Z = X + Y$$

$X, Y \sim U(0,10)$ and indep.

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_{Z|X}(z|x) dx$$

$$f_{XZ}(x,z) = f_X(x) f_{Z|X}(z|x)$$

$$f_Z(z) = \int_{-\infty}^{+\infty} f_{XZ}(x,z) dx = \int_{-\infty}^{+\infty} f_X(x) f_{Z|X}(z|x) dx$$

$$F_Z(z) = P[Z < z] = \int_{-\infty}^{+\infty} f_X(x) P[Z < z | X=x] dx$$

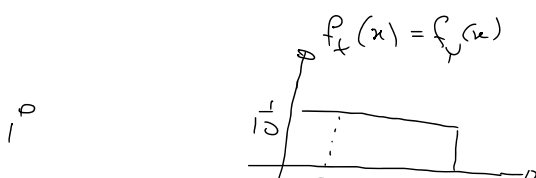
$$f_X(x) P[A|X=x] = P[A] f_{X|A}(x) \Rightarrow P(A) = \int_{-\infty}^{+\infty} f_X(x) P[A|X=x] dx$$

$$Z = X + Y$$

$$Z_{|X=x} = Y + x \quad \begin{matrix} \text{indep} \\ \downarrow \\ Y + x \end{matrix}$$

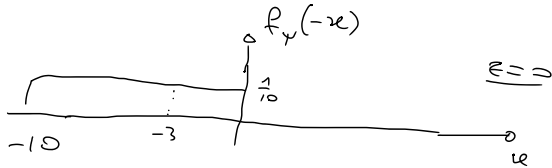
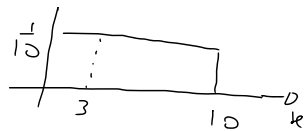
$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_{Z|X}(z|x) dx = \int_{-\infty}^{+\infty} f_X(x) f_{Y+X}(z|x) dx = \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x) dx$$

$$= \int_{-\infty}^{+\infty} f_X(x) f_{Y+X|X=x}(z|x) dx = \int_{-\infty}^{+\infty} f_X(x) f_{Y|X=x}(z-x) dx$$



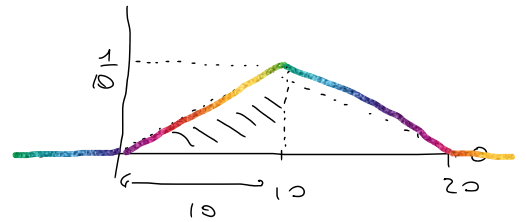
$$= \begin{cases} 0 & z \leq 0 \\ \frac{z}{100} & 0 < z < 10 \\ \frac{20-z}{100} & 10 < z < 20 \\ 0 & z \geq 20 \end{cases}$$

1°



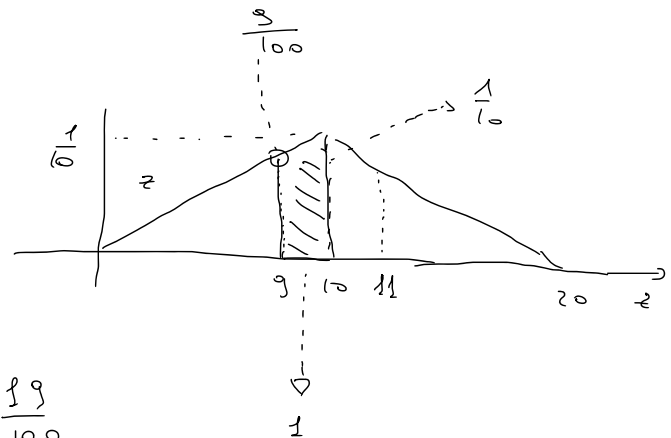
$$z \leq 0$$

$$z \begin{cases} \frac{1}{100} & 0 < z < 10 \\ \frac{20-z}{100} & 10 < z < 20 \\ 0 & z > 20 \end{cases}$$



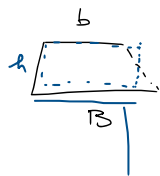
$$\frac{1}{2} \times 10 \times \frac{1}{10} = \frac{1}{2}$$

$$\frac{1}{2} \times 20 \times \frac{1}{10} = 1$$



$$P[9 < z < 11] = \frac{19}{100}$$

$$\frac{1}{2} \left(\frac{1}{10} + \frac{9}{100} \right) \times 2 = 1 = \frac{19}{2 \times 100}$$



$$\frac{b+h}{2}$$

$$X \sim (0, 1)$$

$$Y \sim (0, 10)$$

PDF X

PDF di Y



$$\begin{aligned} 10x & \rightarrow \\ 10y & \rightarrow \\ \hline \tilde{z} &= 10x + 10y \\ \hline \forall z_i \in (9, 19) & \\ \hline n & \end{aligned}$$

$$X \left\{ \begin{aligned} &\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \\ &\left\{ \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11} \right\} \end{aligned} \right.$$

$$Y \left\{ \begin{aligned} &\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \\ &\left\{ \frac{1}{11}, \dots, \frac{1}{11} \right\} \end{aligned} \right.$$

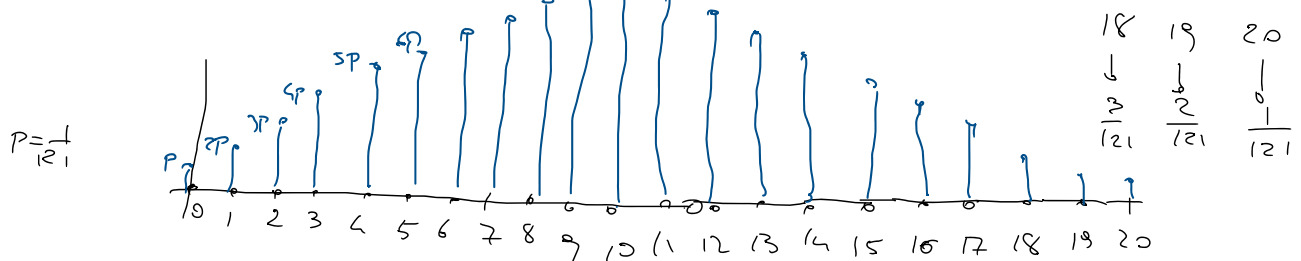
$$\left\{ \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11}, \frac{1}{11} \right\}$$

$$\left\{ \frac{1}{11}, \dots, \frac{1}{11} \right\}$$

$$Z \sim \begin{cases} 0, 1, 2, 3, \dots, 10, \\ 1, 2, 3, \dots, 10, 11, \\ 2, 3, \dots, 10, 11, 12, \\ 3, \dots, 10, 11, 12, 13, \\ 4, 5, \dots, 13, 14, \\ 5, \dots, 15, \\ 6, \dots, 16, \\ 7, \dots, 17, \\ 8, \dots, 18, \\ 9, \dots, 19, \\ 10, \dots, 20 \end{cases}$$

$$\text{Ha prob. } \frac{1}{11^2} = \frac{1}{121}$$

$$\begin{array}{cccccccccccccccccccccccc} \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\} \\ \downarrow \\ \frac{1}{121} \frac{2}{121} \frac{3}{121} \frac{4}{121} \frac{5}{121} \frac{6}{121} \frac{7}{121} \frac{8}{121} \frac{9}{121} \frac{10}{121} \frac{11}{121} \frac{10}{121} \frac{9}{121} \frac{8}{121} \frac{7}{121} \frac{6}{121} \frac{5}{121} \frac{4}{121} \frac{3}{121} \frac{2}{121} \frac{1}{121} \end{array}$$



$$2 \left[\sum_{i=1}^{10} i P \right] + 11 P = 2 P \left[\frac{10 \times 11}{2} \right] + 11 P = P \{ 10 \times 11 + 11 \} = P \{ 121 \} = 1$$

$$P[Z \in \{9, 10, 11\}] = 10P + 11P + 10P = 31P = \frac{31}{121}$$

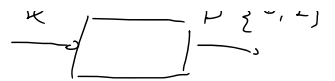
$$P[Z \in \{9, 10\}] = 10P + 11P = 21P = \frac{21}{121} = P[Z \in \{10, 11\}]$$

$$P[9 < Z < 11] = P[Z = 10] = \frac{11}{121} = \frac{1}{11}$$

H_0

$$P\{0, 1\}$$

$$L = \frac{1}{\frac{1}{L} + \frac{1}{L}} = \frac{1}{\frac{2}{L}} = \frac{L}{2}$$



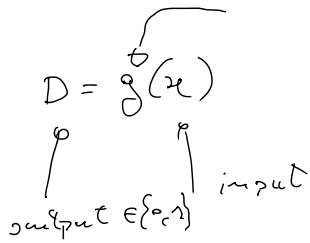
$$X \in \{x_1, x_2, \dots, x_n\}$$

$$H_0: \begin{cases} \{x_1, x_2, \dots, x_n\} \\ \{p_1, p_2, \dots, p_n\} \end{cases}$$

$$H_1: \begin{cases} \{x_1, x_2, \dots, x_n\} \\ \{q_1, q_2, \dots, q_n\} \end{cases}$$

$$P[X=x_i | H_0] = p_i$$

$$P[X=x_i | H_1] = q_i$$



$$\begin{cases} 1, 2, 3 \\ \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \end{cases}$$

$$\begin{cases} 2, 3, 4 \\ \frac{1}{3}, \frac{1}{2}, \frac{1}{6} \end{cases}$$

$$\{1, 2, 3, 4\}$$

$$\left\{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0\right\}$$

$$\left\{0, \frac{1}{3}, \frac{1}{2}, \frac{1}{6}\right\}$$

$$P[D=1 | H_0]$$

$$P[D=0 | H_1]$$

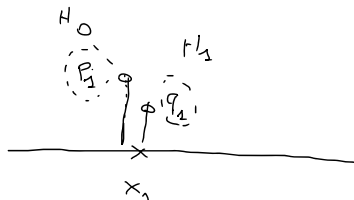
$$P(E) = P(H_0) P(E | H_0) + P(H_1) P(E | H_1)$$



$$= P(H_0) P(D=1 | H_0) + P(H_1) P(D=0 | H_1)$$

$$P(E | X=x_i) = P(H_0) P[E | X=x_i, H_0] + P(H_1) P[E | X=x_i, H_1]$$

$$P(E) = \sum_{i=1}^n P[X=x_i] P[E | X=x_i]$$



$g(x_i) = 0$ zero
solo se vale H_1

$g(x_i) = 1$ uno
solo se vale H_0

$$p_1 \geq q_1$$

$$\begin{aligned} P[X=x_2 | H_2] & \dots p_1 + \dots \\ P[X=x_1 | H_0] & \dots p_1 + \dots \end{aligned}$$

$$P[X=x_1 | H_1] > P[X=x_1 | H_0]$$

regola ML

maximum

likelihood
(verosimiglianza)

verosimiglianza di
 $X=x_1$ nell'ipotesi
 H_0

$$p_1 P(H_0) \quad q_1 P(H_1)$$



maximum likelihood
(verosimiglianza)

$P(H_1)$ a priori (prior)

$$P[H_1 | X=x_1] = \frac{P[H_1] P[X=x_1 | H_1]}{P[X=x_1]} \geq \frac{P[H_0] P[X=x_1 | H_0]}{P[X=x_1]} = P[H_0 | X=x_1]$$

prob. a posteriori